

A 2022 CU Green Labs Fume Hood Sash Contest: Impact on User Behavior, Safety, and Energy Savings at the Jennie Smoly Caruthers Biotechnology Building (JSCBB) at the University of Colorado Boulder



Green Labs Program
UNIVERSITY OF COLORADO **BOULDER**

In partnership with the Environmental Center, Environmental
Health and Safety, and Facilities Management

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Objectives

- Learn about strategies CU Green Labs uses to implement a building-wide contest on lab user behavior change regarding sash closures;
- Learn about the effectiveness of this contest on sash height changes, energy consumption and safety;
- Learn about the incorporation of safety in this contest including the use of posters on fume hood safety that could be beneficial for other research campuses; and
- Learn about plans being explored for continuous fume hood sash monitoring and management.

Background on Fume Hood Sashes

- Closing sashes on variable air volume (VAV) fume hoods is known to benefit energy savings and lab member safety
- Having sashes raised too high can impact air capture efficiency, and potentially harm researcher's health
- On average, 49% to 65% of the hoods are open over a two-week period [1]. Many researchers do not know the importance of closing the sash, not only for energy savings with VAV systems, but also for safety



Courtesy of Kathy Ramirez-Aguilar,
JSCBB Fume Hood

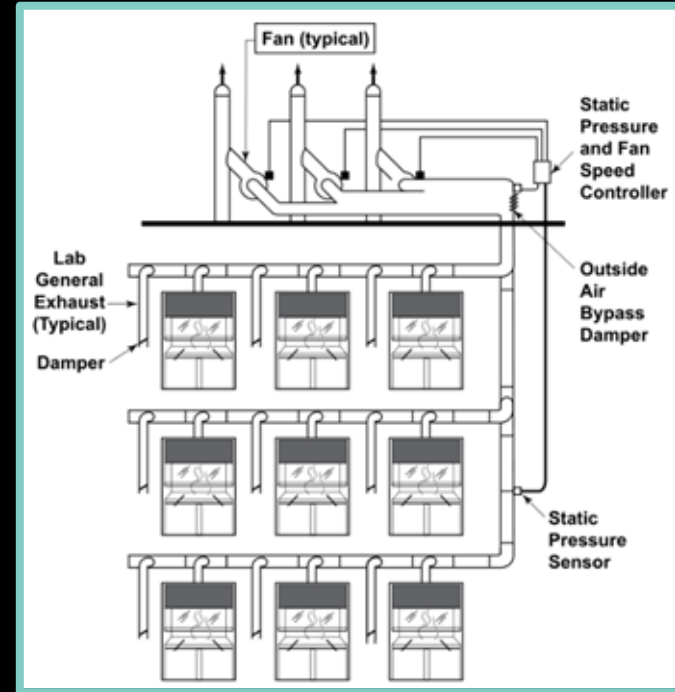
[1] Parker, Scott Michael. "Energy and Cost Savings for Variable Air Volume (VAV) Laboratory Fume Hoods in Two University Research Buildings." *Repository Home*, Carolina State University, 24 Apr. 2006, <https://repository.lib.ncsu.edu/handle/1840.16/2812>.

Background on Door & Ventilation

- There have been ongoing issues with scientists propping lab doors open in JSCBB
- This can affect ventilation performance in the building and even the positive impact of a fume hood sash contest
- Safety concern because the air pressure cascade that must be maintained between spaces in the building
 - there must be a pressure differential so that air from labs must not travel back to hallways

Where Green Labs Comes In

- The CU Green Labs Program has been contributing to fume hood sash management on campus since the program's inception in 2009
- We want to learn how a large-scale fume hood contest impact user behavior, VAV lab building ventilation and energy savings
- Based on the requests from stakeholders on campus, we want to try incorporating door-shutting as an element to a fume hood contest



Courtesy of DELabCon - Environment, Laboratory Safety
October 2, 2019

Before Contest Started...

- Got approval for graphics, messaging, and other details of the contest from stakeholders such as
 - JSCBB Building Management
 - Campus Environmental Health and Safety
 - Facilities Management Engineering and Operations
- We received support and appreciation from all our stakeholders, and list of labs and ~ 200 fume hoods were provided before starting preliminary data collection

CU Green Labs will be visiting your lab for the

JSCBB “JUST SHUT IT” CONTEST FOR FUME HOODS & LAB DOORS

JULY 13 – AUG 3, 2022

Raffle Prizes: Three \$100 gift cards to local restaurants of choice
(The gift cards will go to three different JSCBB labs. A single lab cannot win twice.)

NOTE: If your lab does not want to participate, please email greenlabs@colorado.edu

How the contest works:
During each random visit to your lab, a sticker will be awarded to

1. a fume hood “Just Shut It” contest card if:
 - all panels are closed when hood is not in use (no one standing in front of hood)
 - sash is ≤2 inches or lower when hood is not in use, or ≤14 inches when in use
2. a “Just Close the Door” contest card if:
 - the door between the lab and hallway is not left open and not propped open

CU Green Labs intends to visit your lab ~10 times during the contest. At the end of the contest, each sticker will be converted into a raffle ticket and a raffle drawing will determine the winners.

Background:

- Fume hoods constantly remove conditioned air (heated in the winter or cooled in the summer) from the laboratory and exhaust it from the roof. This air is energy intensive to replace. Thus, while critical to safety in the lab, **fume hoods are likely the largest energy consumer in your lab**. Each fume hood costs approximately the same utilities as a 1500 square foot home for a year. So, you could think of each fume hood being the equivalent of a house's utility bill for a year. JSCBB has energy saving Variable Air Volume (VAV) fume hoods. As a result, closing the sash and panels on your VAV hoods will not only benefit safety in your lab, but also minimize the loss of conditioned air from your lab.
- Closing doors in JSCBB maintains the air pressure cascade between building spaces to ensure that contaminants are contained in the fume hoods and labs. Closing the doors meets this safety intent and facilitates the ventilation system to meet the needs of each space in the most effective and efficient way.

Save energy. Be safe. Please shut your fume hood sash and close your lab door.

This CU Green Labs contest is conducted with the support of Environmental Health and Safety (EHS), Facilities Management (FM), Environmental Center, and JSCBB building management.
For questions or comments, please contact greenlabs@colorado.edu

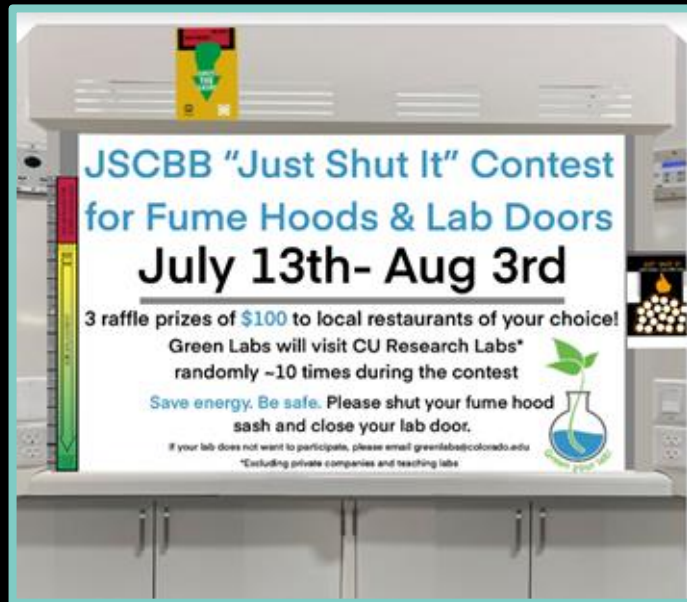
Preliminary Data Collection

- Student Staff Lab Visits
 - Wore lab-appropriate clothing, safety glasses
 - Identifying markers - name tag, Green Labs tag
 - Data collection method was demonstrated to students before hand
 - When asked, say: EH&S project on fume hoods
- Two collections, three weeks apart
 - May 31st, 2022
 - June 20th, 2022 - three weeks before contest



Contest Advertisement & Consent

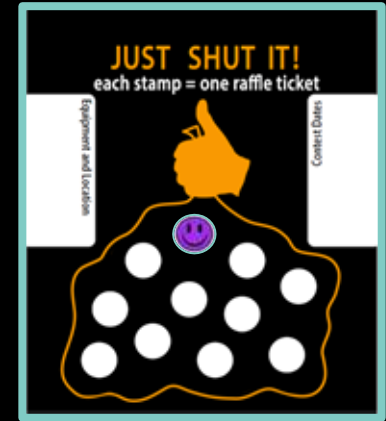
- Flyers
 - Taped to Green Labs poster holders in hallways and rest-areas
 - Simplified version on TV screens in the building
 - Participating lab doors (no companies)
- Community Efforts
 - Email sent out by building management
 - Door-to-door outreach
- Participant Consent
 - Flyers offer an option to opt-out of the contest
 - No lab opted-out of the contest



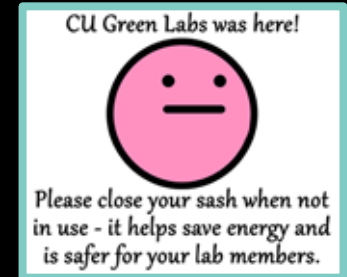
Courtesy of Nick Alter, CU Green Labs Assistant

Data Collection & Incentive System

- Nine data points total, in the span of three weeks
- In addition to collecting data, student assistants
 - Left a smiley face on the contest card when sash is below 2 inches
 - Left a pink-sad-face slip on their fume hood glass if they left the sash open more than 14 inches
 - Left a smiley face on lab door contest card when closed
- The number of smiley faces on contest cards were counted per lab and entered into drawing for three \$100 gift cards for local restaurants of the lab's choice during the final event



Courtesy of University of Toronto



Courtesy: Jackson Kistler,
former CU Green Labs Assistant

Final Event - Ice Cream Social

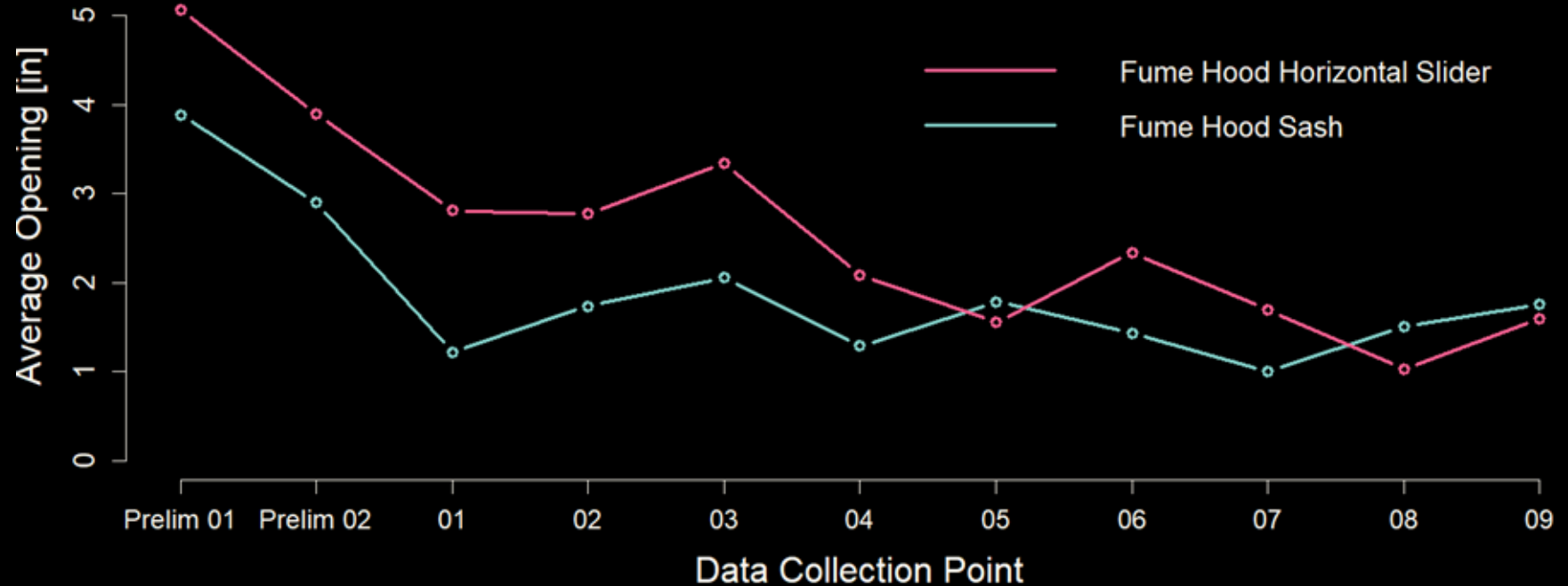
- 75+ participants with overwhelmingly positive feedback
- Raffles drawings
- Thank everyone for their participation in the contest and announcements for what to come in the future



CU Green Labs Team, JSCBB Ice Cream Social
August 8th, 2022

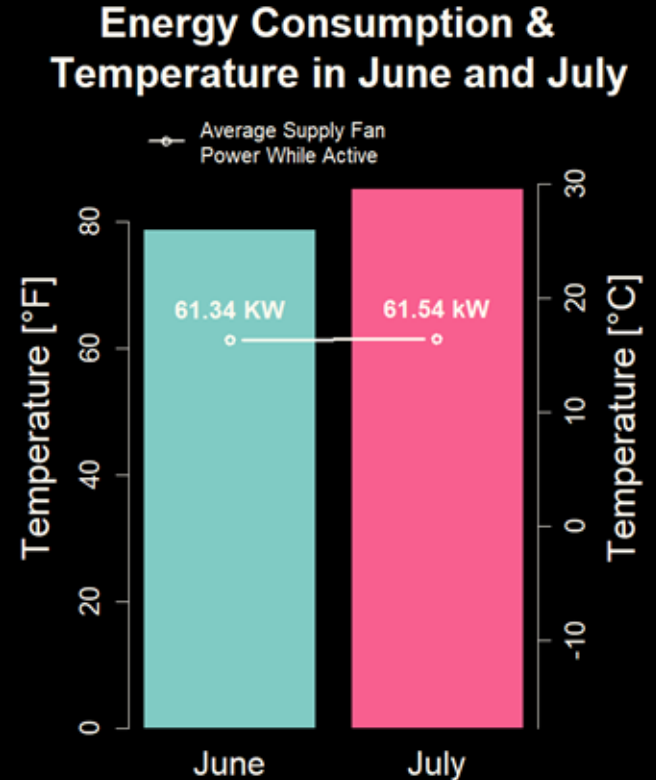
User Behavior Data

**Average Fume Hood Slide Gap and Sash Height
with Each Data Collection**



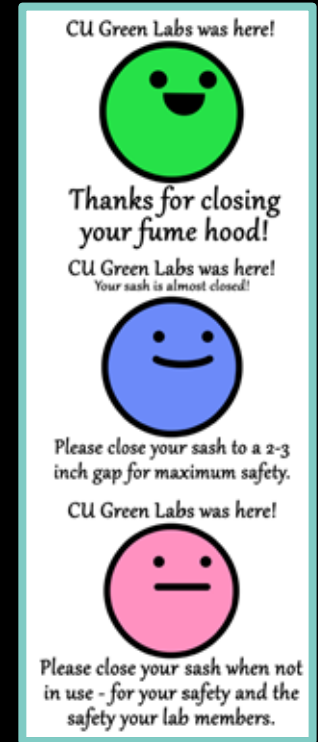
Energy & Other Benefits

- Average supply fan power while active
 - Preliminary Data: June 61.34 kW
 - During & After Contest: July 61.54 kW
- Energy is related to cooling as well, July is 12°F hotter than June
- Unquantifiable Benefits
 - health benefits ~ sash height
 - Educational benefits



Discussion

- Improvements
 - Use full range of sash-glass smileys to improve effectiveness
 - Collecting data weeks after the contest to assess long-term impact of the contest
 - Some complaints that bigger labs with more hoods get more chances to win
- Future plans
 - Implement similar contests in other areas of campus
 - Install fume hood monitors to collect continuous data on sash height and warns people when sash is left open without activity
 - Paper? (hopefully May 2023)



Courtesy of Jackson Kistler

Thank you!

- CU Green Labs
 - Student Assistants: Adam Spicer, Ellie Lindrooth, Nick Alter
 - Program Assistant: Leoncio Lagarde
 - Program Manager: Kathy Ramirez-Aguilar
- CU Campus Partners
 - Facilities Management and EH&S
 - Campus Mechanical Engineer: Shannon Horn & Matt Gess
 - JSCBB Building Management
 - All JSCBB Scientists and Lab Personnel!



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Questions?